

Designator	Quantity	Description
C1, C3, C50	3	CAP CER 4700PF 50V X7R 0603
C2	1	CAP CER 3300PF 50V X7R 0603
C4	1	CAP CER 2200PF 50V X7R 0603
C5, C7, C9, C11, C15, C16, C19, C	22	CAP CER 22UF 25V X5R 0805
C6, C8, C10, C12, C13, C14, C21,	47	CAP CER 0.1UF 50V X7R 0603
C17, C18, C22, C23	4	CAP CER 47UF 25V X5R 1206
C31	1	
C33, C34, C40, C54, C65, C68, C7	13	CAP CER 1UF 25V X7R 0603
C35, C36	2	
C37, C55	2	CAP CER 0.1UF 50V X7R 0402
C38	1	CAP CER 4.7UF 16V X5R 0603
C45, C101, C104	3	CAP TANT POLY 100UF 25V 2917
C46	1	CAP CER 10UF 50V X5R 1206
C49	1	CAP CER 100PF 50V X7R 0603
C51, C96, C107, C112	4	CAP TANT POLY 22UF 25V 2312
C56, C57	2	CAP CER 22PF 50V C0G/NP0 0603
C58, C60, C70, C71	4	CAP CER 12PF 50V C0G/NP0 0603
C61	1	CAP CER 10UF 25V X5R 0603
C102	1	CAP CER 10UF 35V X5R 0805
D1, D2, D10	3	TVS DIODE 16VWM 26VC DO214AA
D3, D4, D5, D6, D7, D8, D9	7	LED BLUE CLEAR CHIP SMD
H1, H2	2	CONN SOCKET 52POS 0.1 GOLD PCB
H3, H4, H5, H6	4	
JTAG1, P1	2	
L1, L2	2	FIXED IND 8.2UH 3A 106 MOHM SMD
L3	1	FIXED IND 4.7UH 5.9A 36.3 MOHM
L4	1	FERRITE BEAD 600 OHM 0603 1LN
NT1	1	
P2	1	CONN FFC FPC 4POS 0.50MM SMD RA
P3	1	CONN FFC FPC 4POS 0.50MM SMD RA
P4	1	
P5	1	
P6, P9	2	CONN FFC FPC 6POS 0.50MM SMD RA
P7	1	
P8	1	
P10	1	
Q1	1	MOSFET N-CH 60V 320MA SOT23-3
R1, R2	2	RES 105K OHM 1% 1/10W 0603
R3, R4, R9, R10, R11, R26, R31, R3	20	RES 10K OHM 1% 1/10W 0603
R5, R6	2	RES 3.32K OHM 1% 1/10W 0603
R7	1	RES 665 OHM 1% 1/10W 0603
R8	1	RES 1.33K OHM 1% 1/10W 0603
R12, R13	2	RES 100K OHM 1% 1/10W 0603

Designator	Quantity	Description
R14, R15	2	RES 20K OHM 1% 1/10W 0603
R16, R17	2	RES 24.9K OHM 1% 1/10W 0603
R18, R19	2	
R20	1	2T WPC2512 3W 50PPM 20 MOHM 1.0%
R21, R24	2	RES 0.01 OHM 1% 1/3W 0603
R22	1	RES 102K OHM 1% 1/10W 0603
R23	1	RES 332K OHM 1% 1/10W 0603
R25	1	RES 43K OHM 1% 1/10W 0603
R27	1	RES 49.9K OHM 1% 1/10W 0603
R28	1	
R29	1	RES SMD 1.3K OHM 1% 1/10W 0603
R30, R33, R34, R35, R36, R43, R44	9	RES SMD 4.7K OHM 1% 1/10W 0603
R37	1	RES SMD 120 OHM 1% 1/10W 0603
R38, R39, R40, R41, R42, R73, R74	9	RES 22 OHM 1% 1/16W 0402
R48	1	RES 120 OHM 1% 1/8W 0805
R50, R51, R53, R55	4	RES 0.01 OHM 1% 3/4W 1206
R56	1	RES 1K OHM 1% 1/10W 0603
R60	1	RES 22 OHM 1% 1/10W 0603
R66, R69	2	RES 5.1K OHM 1% 1/10W 0603
R67, R68, R70, R71, R72	5	RES 1.5K OHM 1% 1/10W 0603
SW1	1	TACTILE SWITCH SPST-NO 0.05A 12V
U1, U2	2	IC ELECTRONIC FUSE 10VQFN
U3, U4	2	IC REG BUCK ADJ 3A 8SOPWR
U5	1	IC REG BUCK BOOST PROG 4A 21VQFN
U6	1	IC MCU 8BIT 128KB FLASH 64TQFP
U7	1	IC CAN CONTROLLER W/SPI 18SOIC
U8	1	IC TRANSCEIVER HALF 1/1 8SOIC
U9	1	IC TRANSCEIVER HALF 1/1 8SOIC
U10	1	IC CURRENT MONITOR 1% 16UQFN
U11	1	SENSOR ANALOG -40C-125C SOT23-3
U12	1	IC TRANSCEIVER FULL 2/2 16TSSOP
U13	1	IC OSC PROG TIMER SOT-23-6
U14	1	IC EEPROM 4KBIT I2C 400KHZ 8SOIC
U15	1	IC REDRIVER I2C HOTSWAP 8TSSOP
X1	1	CRYSTAL 16.0000MHZ 8PF SMD
X2	1	2.0MM X 1.2MM TUNING FORK 32.768
X3	1	CRYSTAL 8.0000MHZ 8PF SMD

Name	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1
4.7nF	Samsung Electro-Me	CL10B472KB8NNNC	Volume Production
3.3nF	Samsung Electro-Me	CL10B332KB8NNNC	Volume Production
2.2nF	Samsung Electro-Me	CL10B222KB8NNNC	Volume Production
22uF	KYOCERA AVX	KGM21AR51E226MU	Volume Production
100nF	Yageo Group	C0603C104K5RACTU	Volume Production
47uF	TDK	C3216X5R1E476M160AC	Volume Production
DNP/100nF			
1uF	TDK	C1608X7R1E105K080AE	Volume Production
DNP/100nF			
100nF	Samsung Electro-Me	CL05B104KB5NNNC	Volume Production
4.7uF	TDK	C1608X5R1C475K080AC	Volume Production
100uF	Vishay	T55D107M025C0060	Volume Production
10uF	TDK	CGA5L3X5R1H106M160AB	Volume Production
100pF	Yageo Group	C0603C101K5RACTU	Volume Production
22uF	KYOCERA AVX	TCJC226M025R0100	Volume Production
22pF	Yageo Group	C0603C220G5GACTU	Volume Production
12pF	Yageo Group	C0603C120J5GACTU	Volume Production
10uF	KYOCERA AVX	06033D106KAT2A	Volume Production
10uF	TDK	C2012X5R1V106K125AC	Volume Production
SMBJ16A	Taiwan Semiconduct	SMBJ16A	Volume Production
Led	Diodes Inc.	LTST-C191TBKT	Volume Production
ESQ-126-14-G-D	Samtec	ESQ-126-14-G-D-LL	Volume Production
Shield			
Header 4			
8.2uH	Vishay	IHLP2525BDER8R2M01	Volume Production
4.7uH	Eaton	EXL1V0503-4R7-R	Volume Production
FB 600R	Murata	BLM18KG601SN1D	Volume Production
NT			
UART_DEBUG	Molex	53261-0471	Volume Production
UART_CONSOLE	Molex	53261-0471	Volume Production
+12V IN			
+12V TEST			
532610671	Molex	53261-0671	Volume Production
+VOUT TEST			
+5V_TRANS TEST			
+5V TEST			
2N7002	onsemi	2N7002KT1G	Volume Production
105k	Yageo Group	RC0603FR-07105KL	Volume Production
10k	Yageo Group	RC0603FR-0710KL	Volume Production
3.32k	Yageo Group	RC0603FR-073K32L	Volume Production
665R	Yageo Group	RC0603FR-07665RL	Volume Production
1.33k	Yageo Group	RC0603FR-071K33L	Volume Production
100k	Yageo Group	RC0603FR-07100KL	Volume Production

Name	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1
20k	Yageo Group	RC0603FR-1320KL	Volume Production
24.9k	Yageo Group	RC0603FR-0724K9L	Volume Production
DNP/4.7k			
20mR	Littelfuse	WPC2512QLR020FYR	Volume Production
10mR	SEI	CSM0603FT10L0	Volume Production
102k	Yageo Group	RC0603FR-07102KL	Volume Production
332k	Yageo Group	RC0603FR-07332KL	Volume Production
43k	Yageo Group	RC0603FR-1343KL	Volume Production
49.9k	Yageo Group	RC0603FR-0749K9L	Volume Production
DNP/150k			
11.3k	Bourns	CR0603-FX-1301ELF	Volume Production
4.7k	Bourns	CR0603-FX-4701ELF	Volume Production
120R	Bourns	CR0603-FX-1200ELF	Volume Production
22R	Yageo Group	RC0402FR-0722RL	Volume Production
120R	Yageo Group	RC0805FR-07120RL	Volume Production
Shunt 4 Pin	Susumu	KRL1632E-M-R010-F-T5	Volume Production
1k	Yageo Group	RC0603FR-071KL	Volume Production
22R	Yageo Group	RC0603FR-0722RL	Volume Production
5.1k	Yageo Group	RC0603FR-075K1L	Volume Production
1.5k	Yageo Group	RC0603FR-071K5L	Volume Production
SW2	C&K Components	PTS636SK25FSMTRLFS	Volume Production
TPS259470ARPWR	Texas Instruments	TPS259470ARPWR	Volume Production
LMR33630ADDA	Texas Instruments	LMR33630ADDA	Volume Production
TPS55287RYQR	Texas Instruments	TPS55287RYQR	Volume Production
ATMEGA128A-AU	Atmel	ATMEGA128AAU	Volume Production
MCP2515-SO	Microchip	MCP2515T-E/SO	Volume Production
SN65HVD251DR	Texas Instruments	SN65HVD251DR	Volume Production
MAX485ESA+	Analog Devices	MAX485ESA+	Volume Production
PAC1934T-I/JQ	Microchip	PAC1934TI/JQ	Volume Production
TC1047AVNBTR	Microchip	TC1047AVNBTR	Volume Production
SN65C3232PWR	Texas Instruments	SN65C3232PWR	Volume Production
TPL5010QDDCTQ1	Texas Instruments	TPL5010QDDCTQ1	Volume Production
24LC04BT-E/SN	Microchip	24LC04BT-E/SN	Volume Production
PCA9508DP,118	NXP Semiconductors	PCA9508DP,118	Volume Production
16Mhz	NDK	STDCSW5-16M	Unknown
Crystal 2 Pins	KYOCERA AVX	DT2012SB32768H5HPWAA	Volume Production
8Mhz	ECS International	ECS-80-8-33Q-ADS	Volume Production

Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
Mouser	187-CL10B472KB8NNNC	0.16	0.48
Mouser	187-CL10B332KB8NNNC	0.2	0.2
Mouser	187-CL10B222KB8NNNC	0.1	0.1
Mouser	581-KGM21AR51E226MU	0.653	14.37
Mouser	80-C0603C104K5R	0.048	2.26
Mouser	810-C3216X5R1E476M	0.94	3.76
Mouser	810-C1608X7R1E105KAE	0.107	1.39
Mouser	187-CL05B104KB5NNNC	0.14	0.28
Mouser	810-C1608X5R1C475KAC	0.24	0.24
Mouser	74-T55D107M025C0060	2.41	7.23
Mouser	810-CGA5L3X5R1H106MB	0.83	0.83
Mouser	80-C0603C101K5R	0.12	0.12
Mouser	581-TCJC226M025R0100	1.96	7.84
Mouser	80-C0603C220G5G	0.39	0.78
Mouser	80-C0603C120J5G	0.035	0.35
Mouser	581-06033D106KAT2A	0.5	0.5
Newark	C2012X5R1V106K125AC	0.416	0.416
Mouser	821-SMBJ16A	0.61	1.83
Mouser	859-LTST-C191TBKT	0.17	1.7
Arrow Electronics	IHLP2525BDER8R2M01	0.7176	1.44
Arrow Electronics	EXL1V0503-4R7-R	0.3453	0.3453
Mouser	81-BLM18KG601SN1D	0.1	0.1
Mouser	538-53261-0471	0.69	0.69
Mouser	538-53261-0471	0.69	0.69
Newark	53261-0671	0.766	1.53
Newark	2N7002KT1G	0.214	0.214
Newark	RC0603FR-07105KL	0.101	0.202
Mouser	603-RC0603FR-0710KL	0.014	0.28
Newark	RC0603FR-073K32L	0.101	0.202
Mouser	603-RC0603FR-07665RL	0.1	0.1
Mouser	603-RC0603FR-071K33L	0.12	0.12
Mouser	603-RC0603FR-07100KL	0.014	0.14

Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
Mouser	603-RC0603FR-1320KL	0.014	0.14
Mouser	603-RC0603FR-0724K9L	0.014	0.14
Arrow Electronics	WPC2512QLR020FYR	0.2033	0.2033
Mouser	708-CSM0603FT10L0	0.66	1.32
Mouser	603-RC0603FR-07102KL	0.1	0.1
Mouser	603-RC0603FR-07332KL	0.1	0.1
Mouser	603-RC0603FR-0749K9L	0.1	0.1
Mouser	652-CR0603FX-1301ELF	0.1	0.1
Mouser	652-CR0603FX-4701ELF	0.012	0.12
Newark	CR0603-FX-1200ELF	0.1	0.1
Mouser	603-RC0402FR-0722RL	0.009	0.09
Mouser	603-RC0805FR-07120RL	0.1	0.1
DigiKey	408-1570-1-ND	0.43	1.72
Mouser	603-RC0603FR-071KL	0.1	0.1
Mouser	603-RC0603FR-0722RL	0.1	0.1
Newark	RC0603FR-075K1L	0.101	0.202
Mouser	603-RC0603FR-071K5L	0.014	0.14
Mouser	595-TPS259470ARPWR	1.72	3.44
Arrow Electronics	LMR33630ADDA	2.88	5.77
Mouser	595-TPS55287RYQR	5.32	5.32
Newark	ATMEGA128A-AU	7.08	7.08
Mouser	579-MCP2515T-E/SO	2.97	2.97
Newark	SN65HVD251DR	4.15	4.15
Newark	MAX485ESA+	6.41	6.41
Arrow Electronics	PAC1934T-I/JQ	1.88	1.88
Arrow Electronics	TC1047AVNBTR	0.7759	0.7759
Mouser	595-SN65C3232PWR	3.31	3.31
Mouser	595-TPL5010QDDCTQ1	1.79	1.79
Arrow Electronics	24LC04BT-E/SN	0.4058	0.4058
Newark	PCA9508DP,118	2.32	2.32
Mouser	344-STDCSW5-16M	1	1
Mouser	581-DT2012SB32768H5H	0.63	0.63
Mouser	520-ECS-80-8-33Q-ADS	0.87	0.87